

3. In a study of physical fitness levels, experts analysed a survey of adults in which they were asked how much they had exercised in the previous four weeks.

The report of the survey, which involved 1615 people, claimed that 30% of the British population did no exercise.

Reported in DailyTelegraph 05/06/15

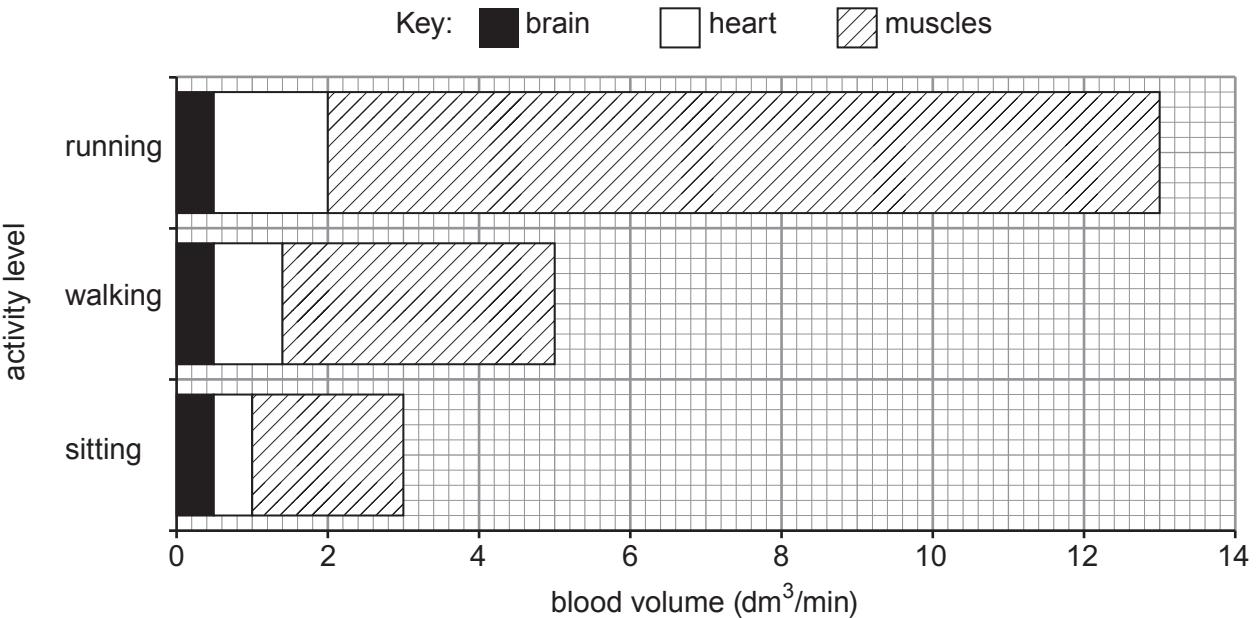
(a) Using the above information, calculate the number of people in the survey who did no exercise. [2]

Number of people =

(b) Why was it important to use a large number of people in this survey? [1]

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(c) The bar chart shows the blood volume (dm³/min) supplied to some parts of the body at rest and during exercise.



Examiner
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(i) Calculate the blood volume supplied to the muscles when sitting. [1]

Blood volume = dm³/min

(ii) Calculate the increase in blood volume supplied to the muscles when running compared to sitting. [1]

Increase in blood volume = dm³/min

(iii) Complete the following sentence. [1]

As blood flows through muscles, substances pass out of narrow blood vessels.

These blood vessels are called

(iv) Three of the following statements correctly describe the result of **an increase in blood volume** supplied to muscles. [3]

Underline the **three** correct statements.

More oxygen is supplied

More lactic acid is produced

More glucose is supplied

More anaerobic respiration takes place

More aerobic respiration takes place

(d) Explain why lack of exercise can result in obesity. [2]

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